

Phase 6: Full Stack Development (FSD) Project Documentation

Project Title: shopEZ — MERN Stack E-Commerce Marketplace

Date: June 2026

Tech Stack: MongoDB - Express.js - React.js - Node.js

1. Project Overview

Purpose

shopEZ is a fully functional, secure, and scalable full-stack e-commerce marketplace platform that delivers a seamless online shopping experience for customers and a powerful administrative control center for store managers and sellers.

Key Features at a Glance

Feature Area	Highlights
Authentication	JWT-based sessions, Bcrypt password hashing, Role-Based Access Control (Customer / Seller / Admin)
Product Catalog	Live autocomplete search, multi-parameter filters (price, category, rating, stock), sort options
Shopping Cart	Persistent cross-session cart, stock validation, real-time totals
Wishlist	Save favorites, one-click "Move to Cart" with auto-clear
Checkout	Multi-step wizard (Address -> Shipping -> Coupon -> Payment), itemized billing
Payments	Mock Stripe Payment Intent endpoint mimicking production-grade integration
Reviews	Verified purchase badge, star ratings, helpful upvote tallies
Order Tracking	Visual 6-step fulfillment timeline (Placed -> Confirmed -> Packed -> Shipped -> Out for Delivery -> Delivered)

Admin Dashboard	Chart.js analytics, product/user/order/coupon CRUD, low-stock alerts
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2. Architecture Overview

shopEZ implements the **Model-View-Controller (MVC)** pattern across a fully decoupled MERN stack:



3. Technology Stack

Layer	Technology	Version	Justification
Frontend Framework	React.js	19.x	Component-driven SPA with virtual DOM for blazing-fast UI updates
Build Tooling	Vite	8.x	Lightning-fast HMR development server with optimized production builds
Styling	Tailwind CSS	3.x	Utility-first responsive design with custom HSL color palettes
Routing (FE)	React Router DOM	6.x	Declarative route hierarchy with protected route guards

HTTP Client	Axios	1.7.x	Promise-based HTTP client with interceptors for auth headers
Charts	Chart.js + react-chartjs-2	4.x	Canvas-rendered bar/doughnut/line charts for admin analytics
Icons	Lucide React	0.395.x	Consistent, accessible SVG icon library
Backend Runtime	Node.js	18+	Server-side JavaScript enabling full-stack JS development
Web Framework	Express.js	4.x	Minimal, unopinionated RESTful API framework
Database	MongoDB + Mongoose	8.x	NoSQL document store with schema validation and ODM
Authentication	JWT + Bcrypt.js	—	Stateless token sessions + secure password hashing
Image Storage	Local uploads (Multer)	1.4.x	Disk-based image persistence with static file serving
Dev Fallback DB	mongodb-memory-server	11.x	In-memory MongoDB for development without external DB config
Dev Tools	Nodemon	3.x	Auto-restart server on file changes

4. Prerequisites

Before running shopEZ locally, ensure the following are installed:

- **Node.js** v18 or above — nodejs.org
- **npm** v8+ (bundled with Node.js)
- **MongoDB** (Local Instance OR MongoDB Atlas cloud cluster)

- **Git** for version control

5. Installation & Setup Guide

Step 1 — Clone the Repository

```
git clone https://github.com/<your-username>/E-Commerce-Application.git
cd "E-Commerce Application"
```

Step 2 — Configure Backend Environment Variables

Navigate to the `backend/` directory and create a `.env` file from the template:

```
cd backend
copy .env.example .env
```

Open `.env` and fill in your values:

```
PORT=5000
MONGO_URI=mongodb://127.0.0.1:27017/shopEZ
JWT_SECRET=your_super_secret_jwt_key_here
STRIPE_API_KEY=your_stripe_api_key_here
```

Note: If `MONGO_URI` is unreachable, the seeder automatically starts a local `mongodb-memory-server` in-memory fallback and seeds it with sample customers, products, reviews, and active coupons.

Step 3 — Install Backend Dependencies

```
# Inside /backend
npm install
```

Step 4 — Install Frontend Dependencies

```
cd ../frontend
npm install
```

Step 5 — (Optional) Seed the Database

```
cd ../backend
npm run seed
```

Seeds the database with mock: admin account, customer accounts, 20+ products across multiple categories, active promotional coupon codes, and sample reviews.

6. Running the Application

Development Mode (Full Stack)

Terminal 1 — Backend API Server:

```
cd backend/  
npm run dev  
# Server starts on http://localhost:5000
```

Terminal 2 — Frontend Dev Server:

```
cd frontend/  
npm run dev  
# App starts on http://localhost:5173
```

7. Folder Structure

Backend (/backend)

```
backend/  
+-- config/  
|   +-- db.js           # MongoDB connection (with in-memory fallback)  
+-- controllers/  
|   +-- authController.js # User register, login, profile, wishlist, addresses  
|   +-- productController.js # Catalog search, reviews, helpful votes  
|   +-- orderController.js # Order creation, payment update, delivery milestones  
|   +-- couponController.js # Coupon CRUD, active validation  
|   +-- analyticsRoutes.js # Sales charts & revenue aggregation  
+-- middleware/  
|   +-- authMiddleware.js # JWT protect(), admin(), seller() guards  
|   +-- errorMiddleware.js # Centralized notFound & errorHandler  
+-- models/  
|   +-- User.js          # User schema (addresses[], wishlist[], role)  
|   +-- Product.js       # Product schema (reviews[], stock, rating)  
|   +-- Order.js         # Order schema (items, payment, delivery timeline)  
|   +-- Review.js        # Review schema (verified, helpfulVotes)  
|   +-- Coupon.js        # Coupon schema (code, type, value, expiry)  
|   +-- Seller.js        # Seller profile schema  
+-- routes/  
|   +-- authRoutes.js    # /api/users (register, login, profile, addresses)
```

```
| |-- productRoutes.js # /api/products (list, CRUD, reviews, suggestions)
| |-- orderRoutes.js # /api/orders (create, pay, deliver)
| |-- couponRoutes.js # /api/coupons (admin CRUD, user validate)
| |-- uploadRoutes.js # /api/upload (Multer file upload)
| |-- analyticsRoutes.js # /api/analytics (sales metrics)
|-- utils/
| |-- seeder.js # Database seed script
|-- uploads/ # Local product image storage
|-- server.js # Express app entry point
|-- .env.example # Environment variable template
```

Frontend (/frontend)

```
Frontend/
|-- src/
| |-- assets/ # Static assets (logos, icons)
| |-- components/
| | |-- Navbar.jsx # Top navigation with cart badge, auth links
| | |-- ProductCard.jsx # Reusable product listing card
| | |-- ProtectedRoute.jsx # Route guard (auth + role verification)
| |-- context/
| | |-- AuthContext.jsx # User session state & dispatch
| | |-- CartContext.jsx # Shopping cart state (items, totals)
| | |-- ToastContext.jsx # Global toast notification system
| |-- pages/
| | |-- Home.jsx # Catalog browse with search & filters
| | |-- ProductDetails.jsx # Single product, reviews, related carousel
| | |-- Cart.jsx # Cart summary with quantity controls
| | |-- Wishlist.jsx # Saved wishlist with move-to-cart
| | |-- Checkout.jsx # Multi-step wizard with coupon & mock payment
| | |-- MyOrders.jsx # Order history with fulfillment tracker
| | |-- ProfileDashboard.jsx # Edit profile, manage addresses
| | |-- AdminDashboard.jsx # Admin panel (analytics, CRUD, coupons)
| | |-- SellerDashboard.jsx # Seller panel (products, orders)
| | |-- Login.jsx # Sign-in form
| | |-- Register.jsx # Registration form
| |-- utils/ # Helper functions and Axios config
| |-- App.jsx # Root routes definition
|-- main.jsx # ReactDOM entry point
|-- index.html # Vite HTML template
|-- vite.config.js # Vite configuration
```

8. RESTful API Documentation

All API routes are served from <http://localhost:5000>. Protected routes require a valid [Authorization: Bearer <jwt_token>](#) header.

8.1 Authentication Routes — /api/users

Method	Endpoint	Auth	Description
POST	/api/users	Public	Register a new user account
POST	/api/users/login	Public	Log in and retrieve JWT token
GET	/api/users/profile	User	Get current user profile
PUT	/api/users/profile	User	Update name, email, or password
GET	/api/users/wishlist	User	Retrieve user's wishlist products
POST	/api/users/wishlist/:productId	User	Toggle product in/out of wishlist
POST	/api/users/addresses	User	Add a new shipping address
PUT	/api/users/addresses/:addressId	User	Edit an existing shipping address
DELETE	/api/users/addresses/:addressId	User	Remove a shipping address
GET	/api/users	Admin	Get all registered users
PUT	/api/users/:id/block	Admin	Block or unblock a user account

8.2 Product Routes — /api/products

Method	Endpoint	Auth	Description
GET	/api/products	Public	List all products with optional search/filter/sort
GET	/api/products/suggestions	Public	Get autocomplete name suggestions
GET	/api/products/:id	Public	Get single product detail

GET	/api/products/:id/related	Public	Get related product recommendations
POST	/api/products	Seller/Admin	Create a new product listing
PUT	/api/products/:id	Seller/Admin	Update product information
DELETE	/api/products/:id	Seller/Admin	Delete a product listing
POST	/api/products/:id/reviews	User	Submit a product review with star rating
POST	/api/products/:id/reviews/:reviewId/vote	User	Toggle helpful vote on a review

8.3 Order Routes — /api/orders

Method	Endpoint	Auth	Description
POST	/api/orders	User	Place a new order (deducts stock)
GET	/api/orders/myorders	User	Retrieve current user's order history
GET	/api/orders/:id	User	Get single order details
PUT	/api/orders/:id/pay	User	Update order payment status to paid
GET	/api/orders	Admin	List all platform orders
PUT	/api/orders/:id/delivery	Admin	Update order delivery status milestone

8.4 Coupon Routes — /api/coupons

Method	Endpoint	Auth	Description
GET	/api/coupons	Admin	List all coupon codes

POST	/api/coupons	Admin	Create a new coupon rule
PUT	/api/coupons/:id	Admin	Edit coupon details or toggle active status
DELETE	/api/coupons/:id	Admin	Delete a coupon permanently
POST	/api/coupons/validate	User	Validate coupon code and retrieve discount

8.5 Payment Route — /api/payment

Method	Endpoint	Auth	Description
POST	/api/payment/create-intent	User	Create a mock Stripe payment intent returning a client secret

8.6 Upload Route — /api/upload

Method	Endpoint	Auth	Description
POST	/api/upload	Seller/Admin	Upload a product image via Multer (stored in /uploads)

9. Authentication & Security Model

- **Password Hashing:** All user passwords are hashed via `bcryptjs` with a salt round factor of 10 before being persisted to MongoDB.
- **Session Tokens:** Upon successful login, the server signs a JWT containing `user._id` and `user.role` with a configurable secret (`JWT_SECRET`). Tokens carry a 30-day expiry.
- **Protected Routes (Backend):** The `protect` middleware in `authMiddleware.js` verifies the `Authorization` header on every protected endpoint. The `admin()` and `seller()` middleware further restrict access based on the decoded role field.
- **Protected Routes (Frontend):** The `ProtectedRoute.jsx` component wraps sensitive React pages, redirecting unauthenticated users to the Login page.

10. Known Constraints

- **Payment Integration:** The current Stripe integration uses a simulated mock endpoint. A real Stripe publishable key and a configured webhook listener would be required for live transaction processing.
 - **Image Storage:** Product images are stored locally in the [backend/uploads/](#) directory. For production deployments, replacing Multer's disk storage with Cloudinary or AWS S3 integration is recommended.
 - **Seller Profile:** Seller onboarding requires admin approval via the admin panel before a seller account gains product management privileges.
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11. Future Enhancements

- **Real Payment Gateway Integration:** Wire actual Stripe webhook handling to confirm payment events asynchronously.
- **Cloud Image CDN:** Integrate Cloudinary SDK to replace local Multer disk storage.
- **AI Product Recommendations:** Implement ML-driven collaborative filtering to personalize related-product suggestions.
- **Multi-Vendor Marketplace:** Expand seller onboarding to support independent vendor profiles, individual seller pages, and commission tracking.
- **Advanced Analytics:** Add time-series graphs, cohort retention charts, and inventory forecasting to the Admin Dashboard.
- **Push Notifications:** Notify customers via email or browser push on order status milestones (Shipped, Delivered).